CS316

INTRODUCTION TO AI AND DATA SCIENCE

Topic 9

Fundamental Statistical Theorems in Data Science

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1. Bayes theorem
2. Law of Large Numbers (LLN)
3. Central Limit theorem (CLT)

1. BAYES THEOREM

Scenario: Medical diagnosis

probability vs. accuracy (true pos)

depends on de sample

Scenario: You're testing for a rare disease

Let's say:

- Only **1 in 1,000** people have it → $P(\text{Disease})=0.001$
- The test is **99% accurate**:
  - If you **have** the disease: 99% chance the test is positive → $P(+|\text{Disease})=0.99$
  - If you **don't have** it, the probability the test is positive is: 1% (false positive) → $P(+|\text{NoDisease})=0.01$

You take the test and get a **positive result**. Now the big question:

"What are the chances I really have the disease?" : $P(\text{Disease}|+)$??

Bayes' Theorem: Updating Beliefs with Data

You will not be asked abt the bayes rule in this course

- **Definition**: A fundamental result in probability theory that updates the likelihood of an event based on new evidence.
- **Formula**: $P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$
  - Where:
    - $P(A|B)$: Probability of event A given B is true.
    - $P(B|A)$: Probability of event B given A is true.
    - $P(A)$: Probability of event A.
    - $P(B)$: Probability of event B.
- **Significance in Data Science**:
  - **Predictive Modeling**: Enables probabilistic predictions incorporating new data.
  - **Hypothesis Testing**: Fundamental for updating model predictions based on evidence.
  - **Machine Learning**: Underpins many algorithms, especially in areas requiring continual learning from data.

Scenario: Medical diagnosis (ctd.)

Out of **1000 people**:

- Only **1** actually has the disease.
- That person will likely test **positive** (0.99 probability) → So 0.99 **true positives**
- The other **999** don't have it. But 1% of them will test positive → 9.99 **false positives**

So:

- **True positives**: 0.99
- **False positives**: 9.99
- **Total positives**: 0.99 + 9.99 ≈ 10.98 so P(+) ≈ 0.011

Now your **actual chance of having the disease** given a positive result is:

$P(\text{Disease} \mid +) = P(+|\text{Disease}) * P(\text{Disease}) / P(+) = 0.99 * 0.001 / 0.011 \approx 0.09$ or **9%**

Likelihood Estimation

$y = w_1 x_1 + w_2 x_2 + \dots w_n$ (linear regression), minimize the error

$y = P(Y = y_i | X = x_i)$ classification, maximize the likelihood

↳ parameters θ_i

$\max P(\text{data} | \theta)$ find the max using bayes theorem

$y = \theta_1 x_1 + \theta_2 x_2$ (x are features of cat, is it cat or dog) a certain value of x would mean its a cat

start by random values for θ, repeat multiple iterations using the likelihood of the problem ex: if the probability is 0.7 cat ⇒ the pic is a cat

MLE uses **probability** to find the parameter value that **maximizes the likelihood** of the observed data.

It assumes a known model (distribution) and finds the **best-fitting parameters**.

Formula:

$\text{MLE} = \arg_\theta \max P(\text{data}|\theta)$

Think of it as: *"Which parameter value makes this data most probable?"*

Example: Estimating the bias of a coin (θ) from observed heads/tails.

Apriori Estimation (Bayesian Inference)

Bayesian inference adds **prior beliefs** to probability estimation.

Uses **Bayes' Theorem**:

$P(\theta|data) = P(data|\theta) \cdot P(\theta) / P(data)$

Here:

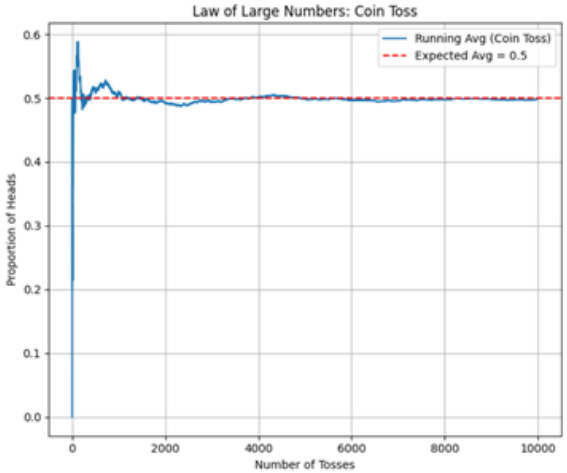
- $P(\theta)$: **Prior** – belief before seeing data
- $P(data|\theta)$: **Likelihood** (same as MLE)
- $P(\theta|data)$: **Posterior** – updated belief
- **Bayesian Estimation** = maximize posterior probability:
- $\theta^{MAP} = \arg\theta \max P(\theta|\ data)$

2. LAW OF LARGE NUMBERS

Scenario 1: Coin tossing

Imagine you're at a local fair, and there's a game booth where you can toss a **fair coin** (i.e., 50% chance heads, 50% chance tails).

You're curious to see if the coin is really fair, so you decide to start flipping it and recording the outcomes.



Small Sample (10 Tosses):

You flip the coin **10 times** and get:

> **Heads: 7**
> **Tails: 3**

That's **70% heads, 30% tails.** That seems off from the expected 50-50, right? But it's a small sample, so some randomness is expected.

not reliable

More Tosses (100 Tosses):

Now you flip the coin **100 times.** This time you get:

> **Heads: 53**
> **Tails: 47**

You're getting **closer to 50%.** Still some variation, but not as extreme as before.

Even More Tosses (10,000 Tosses):

You keep going—now you've flipped the coin **10,000 times.** You observe:

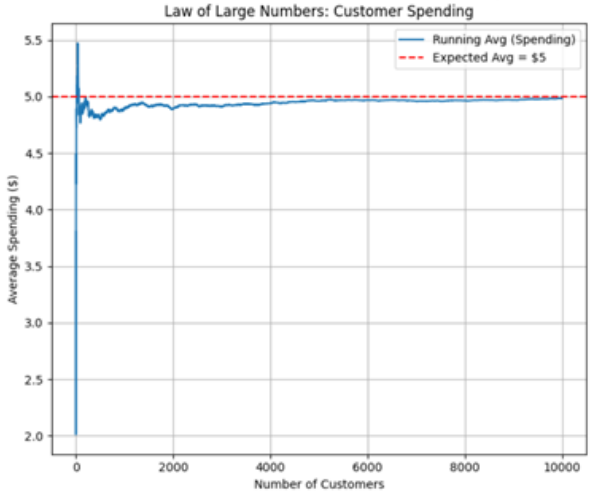
> **Heads: 5,012**
> **Tails: 4,988**

Now you're **very close to 50%.** The percentage of heads is **50.12%**, almost spot on.

more samples ⇒ higher converge to the true probability

Scenario 2 : A Coffee Shop Tracks Customer Spending

Imagine you run a small **coffee shop**, and you're curious about **how much an average customer spends per visit**. You've heard that on average, customers spend **$5**, but you're not entirely convinced, so you decide to test it out.



📊 **Day 1: Small Sample (5 Customers)**

You track the first **5 customers**, and they spend:

> $7, $3, $10, $2, $4
> ➤ Average = ($7 + $3 + $10 + $2 + $4) / 5 = $5.2

That's close to $5, but it still feels a bit random—one big spender skewed it.

📅 **After 1 Week: Medium Sample (100 Customers)**

Now you have data for **100 customers**. The average spending turns out to be:

> ➤ $4.87

That's **even closer** to $5. The fluctuations are calming down.

📅📅 **After 1 Month: Large Sample (3,000 Customers)**

You continue tracking over the month. With 3,000 customers, the average spending becomes:

> ➤ $5.01

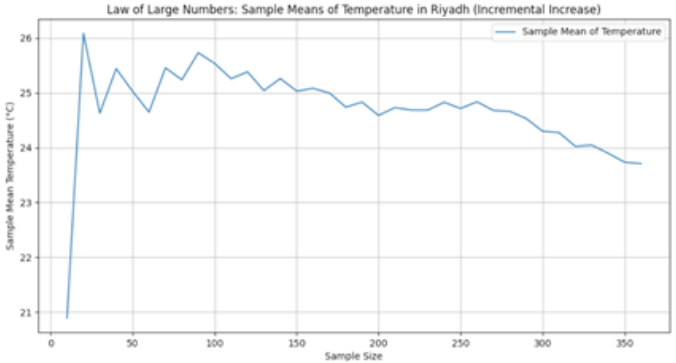
Now you're almost exactly at the expected average. Despite individual variation, the **overall pattern stabilizes**.

Law of Large Numbers

population [cannot have in the data set]
so have an approximation like of the mean by having the largest sample

$$\lim_{n\to\infty} P\left(\left|\bar{X} - \mu\right| > \epsilon\right) = 0$$

- **Definition:** The Law of Large Numbers (LLN) ensures that the average of results from a large number of trials converges to the expected mean as trials increase.
- **Key Points:**
  - **Convergence:** Sample averages approximate the population mean with a large number of trials.
  - **Foundation:** Essential for validating statistical models and results in data science.
  - **Applications:** Underlies inferential statistics, enabling population generalizations from sample data.
- **Significance:** Validates that, given sufficient data, sample statistics can reliably estimate population parameters, critical for data-driven decision-making.

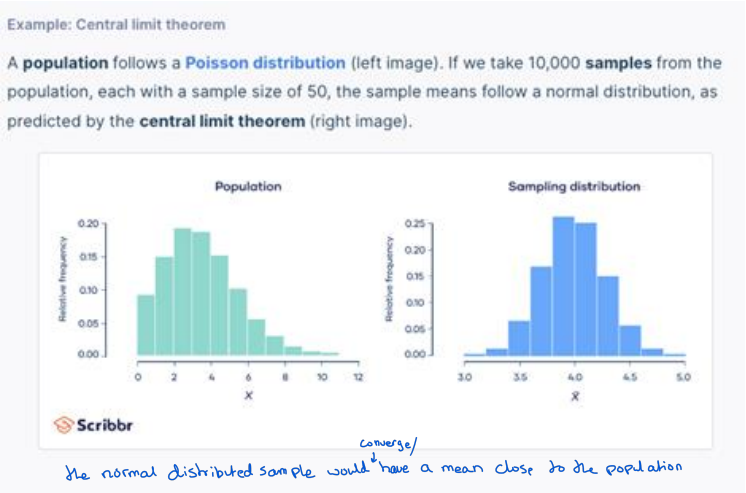


In simple terms:
do something many times → take the average → it gets close to the true mean.

3. CENTRAL LIMIT THEOREM (CLT)

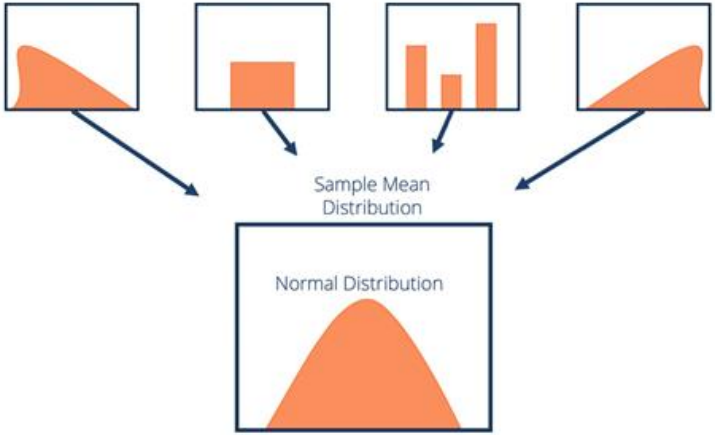
Central Limit Theorem (CLT)

- **Definition:**
  - The CLT states that if you take **sufficiently large, random samples** from a population with a finite mean (μ) and variance (σ^2), the **distribution of the sample means will be approximately normal**.
  - This holds true regardless of the shape of the original population distribution.
  - The larger the sample size, the closer the approximation to a normal distribution.
- **Key Components:**
  - **Large Number of Observations:** The theorem applies when dealing with a sufficiently large sample size.
  - **Finite Mean and Variance:** Each random variable must have a finite mean (μ) and variance (σ_2).
- **Significance in Data Science:**
  - **Justification for Normal Distribution:** Allows the use of normal distribution in statistical methods, even if the original data isn't normally distributed.
  - **Foundation for Statistical Methods:** Vital for hypothesis testing, constructing confidence intervals, and much of the inferential statistics toolkit.



https://www.scribbr.com/statistics/central-limit-theorem/

The Power of Central Limit Theorem (CLT) in Diverse Data Analysis



https://corporatefinanceinstitute.com/resources/data-science/central-limit-theorem/

Scenario: Cookie Weighing

🍪 Cookie Weighing Example

Imagine you run a bakery, and you make chocolate chip cookies every day.

You don't weigh every single cookie, but you know:

- Sometimes the cookies are heavier (more dough, more chips),
- Sometimes lighter (less dough),
- So, cookie weights are **not consistent** and **not normally distributed**—some are way off!

🔍 What do you want to know?

You want to know the **average weight** of your cookies so you can:

- Control portion sizes,
- Price them correctly,
- Stay consistent for customers.

But you don't have time to weigh all 1,000 cookies each day.

✅ So what do you do?

You randomly pick:

- 10 cookies one day → take the average weight.
- 10 cookies the next day → take another average.
- You repeat this every day for a few weeks.

Now you have **a list of averages** (each one from 10 cookies).

💥 Here's where the **Central Limit Theorem** kicks in:

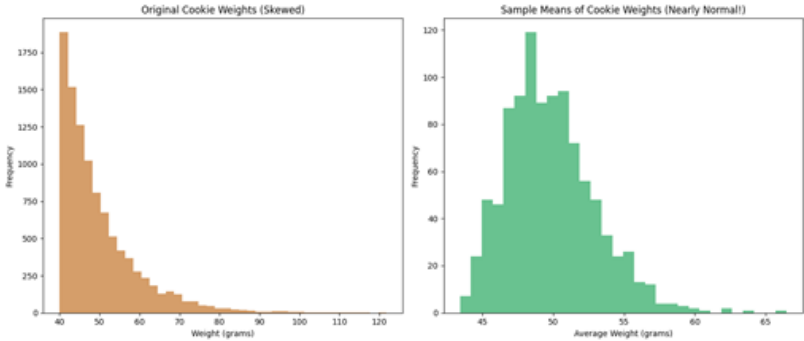
Even though **individual cookie weights** are all over the place, the **averages** you collected:

- Start to form a **normal (bell-shaped) pattern**.
- This means the average of your sample averages is a good estimate of the **true average** cookie weight!

🎯 Why does this matter?

You can now:

- Predict your average cookie weight.
- Know how much variation to expect.
- Make reliable decisions **without needing perfect data.**



Data Entropy

A game of “Guess Who”

Think of information as attributes of the characters.

Information is **high** when the attribute is “rare”

Has a computer is a rare attribute. It directly identifies the character. **High** information gain.

“Is a person” is a common attribute. It identifies all characters. **Low** information gain.

Do you see a relationship forming between probability and information?



What about entropy?

What is Entropy?

- Measures the **expected information**, or **uncertainty** or **surprise** in a probability distribution.
- High entropy → **More uncertainty (randomness)**
- Low entropy → **Less uncertainty (predictable events)**



Summary

- The following important concepts have been covered:
  - The 3 fundamental theorems in statistical learning: their importance and some practical applications
  - Data Entropy

Next Chapter :

  - Hypothesis Testing